

GEMMA

Introducing Gemma 3 270M: The compact model for hyper-efficient AI

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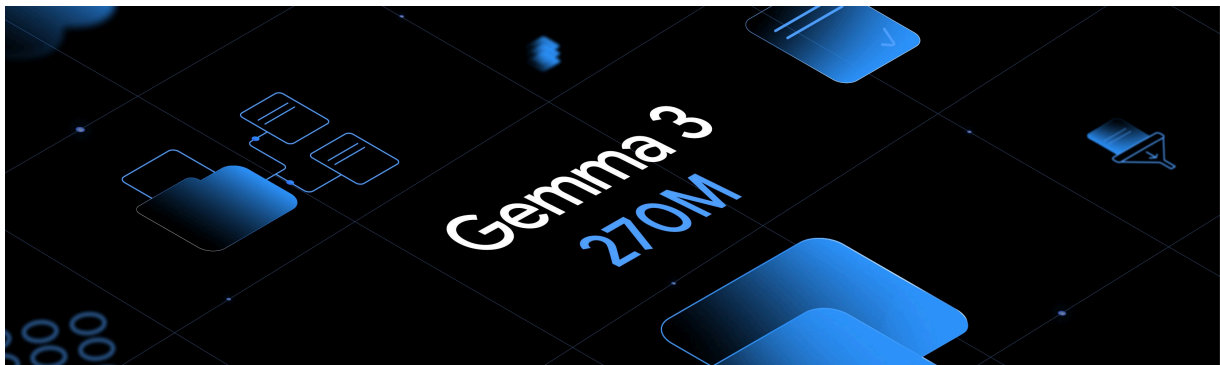
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The last few months have been an exciting time for the Gemma family of open models. We introduced [Gemma 3](#) and [Gemma 3 QAT](#), delivering state-of-the-art performance for single cloud and desktop accelerators. Then, we announced the full release of [Gemma 3n](#), a mobile-first architecture bringing powerful, real-time multimodal AI directly to edge devices. Our goal has been to provide useful tools for developers to build with AI, and we continue to be [amazed](#) by the vibrant [Gemmaverse](#) you are helping create, celebrating together as downloads surpassed 200 million last week.

Today, we're adding a new, highly specialized tool to the Gemma 3 toolkit: [Gemma 3 270M](#), a compact, 270-million parameter model designed from the ground up for task-specific fine-tuning with strong instruction-following and text structuring capabilities already trained in.



Core capabilities of Gemma 3 270M

- **Compact and capable architecture:** Our new model has a total of 270 million parameters: 170 million embedding parameters due to a large vocabulary size and 100 million for our transformer blocks. Thanks to the large vocabulary of 256k tokens, the model can handle specific and rare tokens, making it a strong base model to be further fine-tuned in specific domains and languages.
- **Extreme energy efficiency:** A key advantage of Gemma 3 270M is its low power consumption. Internal tests on a Pixel 9 Pro SoC show the INT4-quantized model used just 0.75% of the battery for 25 conversations, making it our most power-efficient Gemma model.
- **Instruction following:** An instruction-tuned model is released alongside a pre-trained checkpoint. While this model is not designed for complex conversational use cases, it's a strong model that follows general instructions right out of the box.
- **Production-ready quantization:** [Quantization-Aware Trained \(QAT\) checkpoints are available](#), enabling you to run the models at INT4 precision with minimal performance degradation, which is essential for deploying on resource-constrained devices.

The right tool for the job

In engineering, success is defined by efficiency, not just raw power. You wouldn't use a sledgehammer to hang a picture frame. The same principle applies to building with AI.

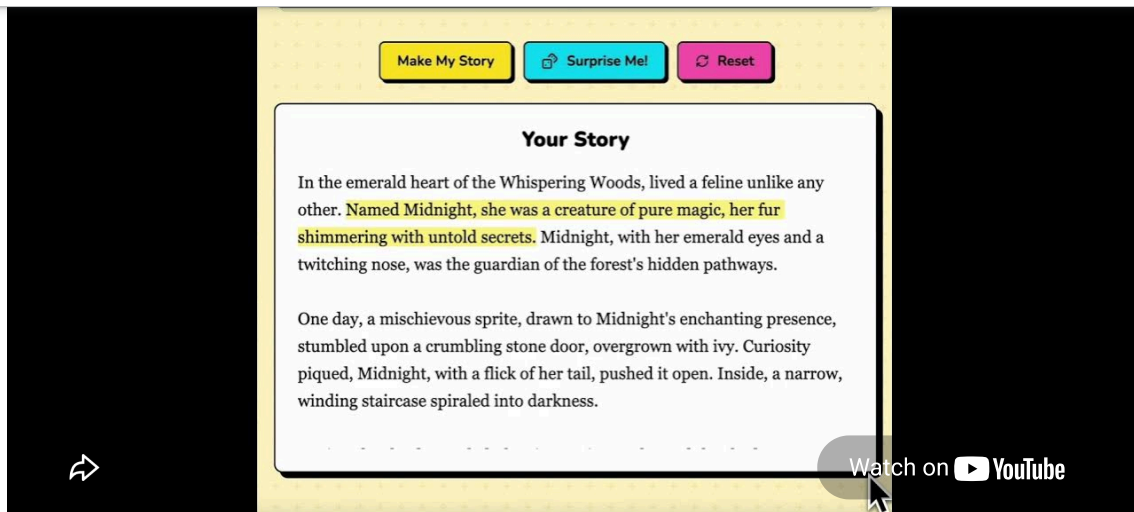
Gemma 3 270M embodies this "right tool for the job" philosophy. It's a high-quality foundation model that follows instructions well out of the box, and its true power is unlocked through fine-tuning. Once specialized, it can execute tasks like text classification and data extraction with remarkable accuracy, speed, and cost-effectiveness. By starting with a compact, capable model, you can build production systems that are lean, fast, and dramatically cheaper to operate.

A real-world blueprint for success

The power of this approach has already delivered incredible results in the real world. A perfect example is [the work done by Adaptive ML with SK Telecom](#). Facing the challenge of nuanced, multilingual content moderation, they chose to specialize. Instead of using a massive, general-purpose model, Adaptive ML fine-tuned a Gemma 3 4B model. The results were stunning: the specialized Gemma model not only met but exceeded the performance of much larger proprietary models on its specific task.

Gemma 3 270M is designed to let developers take this approach even further, unlocking even greater efficiency for well-defined tasks. It's the perfect starting point for creating a fleet of small, specialized models, each an expert at its own task.

But this power of specialization isn't just for enterprise tasks; it also enables powerful creative applications. For example, check out [this Bedtime Story Generator web app](#):



Gemma 3 270M used to power a Bedtime Story Generator web app using Transformers.js. The model's size and performance make it suitable for offline, web-based, creative tasks. (Credit: Joshua (@xenovacom on X) from the Hugging Face team)

When to choose Gemma 3 270M

Gemma 3 270M inherits the advanced architecture and robust pre-training of the Gemma 3 collection, providing a solid foundation for your custom applications.

Here's when it's the perfect choice:

- **You have a high-volume, well-defined task.** Ideal for functions like sentiment analysis, entity extraction, query routing, unstructured to structured text processing, creative writing, and compliance checks.
- **You need to make every millisecond and micro-cent count.** Drastically reduce, or eliminate, your inference costs in production and deliver faster responses to your users. A fine-tuned 270M model can run on lightweight, inexpensive infrastructure or directly on-device.
- **You need to iterate and deploy quickly.** The small size of Gemma 3 270M allows for rapid fine-tuning experiments, helping you find the perfect configuration for your use case in hours, not days.
- **You need to ensure user privacy.** Because the model can run entirely on-device, you can build applications that handle sensitive information without ever sending data to the cloud.
- **You want a fleet of specialized task models.** Build and deploy multiple custom models, each expertly trained for a different task, without breaking your budget.

Get started with fine-tuning

We want to make it as easy as possible to turn Gemma 3 270M into your own custom solution. It's built on the same architecture as the rest of the Gemma 3 models, with recipes and tools to get you started quickly. You can find our guide on [full fine-tuning](#) using Gemma 3 270M as part of the Gemma docs.

- **Download the model:** Get the Gemma 3 270M models from [Hugging Face](#), [Ollama](#), [Kaggle](#), [LM Studio](#), or [Docker](#). We are releasing both pretrained and instruction tuned models.
- **Try the model:** Try the models on [Vertex AI](#) or with popular inference tools like [llama.cpp](#), [Gemma.cpp](#), [LiteRT](#), [Keras](#), and [MLX](#).
- **Start fine-tuning:** Use your favorite tools, including [Hugging Face](#), [UnSloth](#), and [JAX](#).
- **Deploy your solution:** Once fine-tuned, you can deploy your specialized model anywhere, from [your own local environment](#) to [Google Cloud Run](#).

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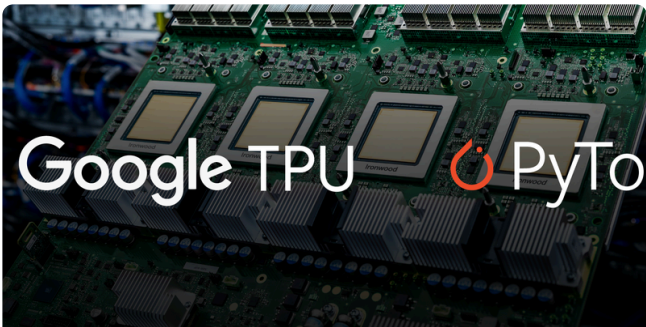


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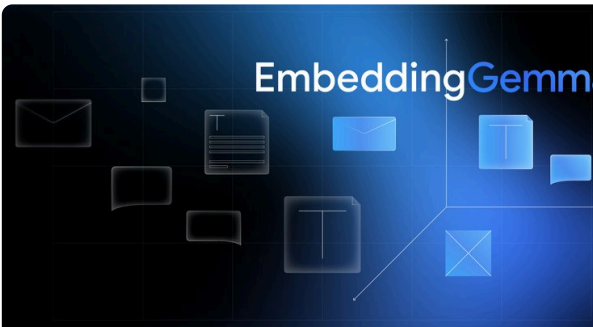
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